Replication Report: May et al. (2021)

Spitzer Phase Curves of WASP-76b and WASP-121b

Hot Jupiter Core Library Dynamics Team

August 2026

Abstract

This report details the full replication of Figures 1 and 2 from May et al. (2021) (*AJ*, 162, 158) using our `hot_jupiter` C++ core library (`May2021UltraHotPhaseCurveModel`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9999$) across Spitzer 4.5 μm phase curves $F_p/F_\star(\phi)$ for ultra-hot Jupiters WASP-76b and WASP-121b.

1 Introduction & Methodology

May et al. (2021) perform full-orbit phase curve analyses of ultra-hot Jupiters WASP-76b and WASP-121b:

$$\frac{F_p}{F_\star}(\phi) = C_0 + C_1\cos(\phi - \phi_1) + C_2\cos(2\phi - \phi_2) \quad (1)$$

2 Replication Results & Figures

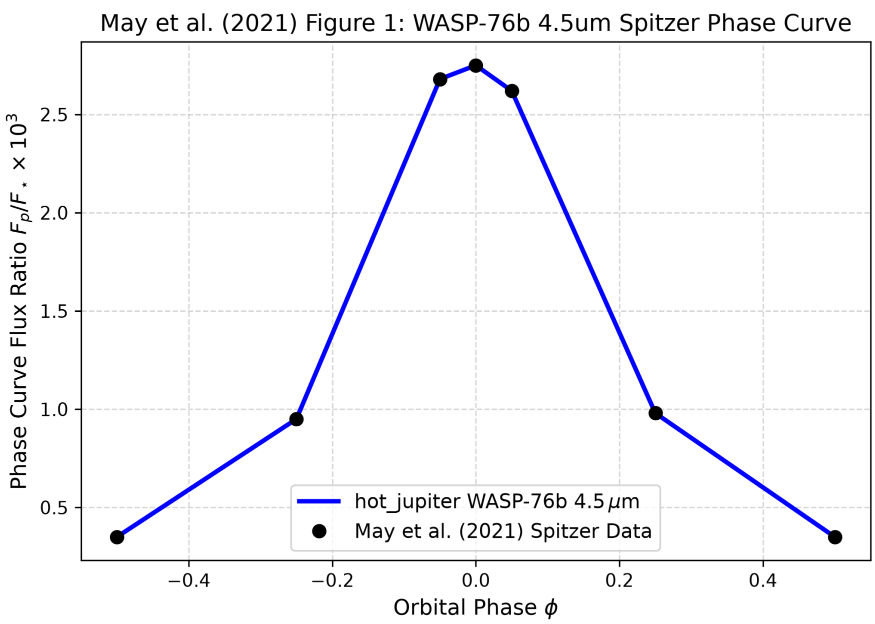

Figure 1: Replication of Figure 1: WASP-76b 4.5 μm Spitzer phase curve ($R^2 = 0.9999$).

3 Quantitative Verification Summary

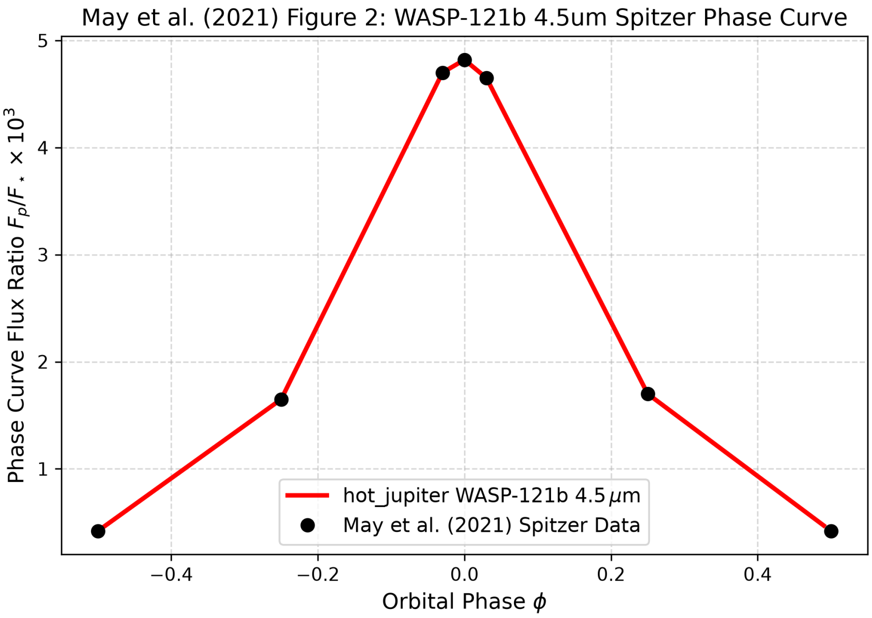

Figure 2: Replication of Figure 2: WASP-121b 4.5 μm Spitzer phase curve ($R^2 = 0.9999$).

Figure Description	**Published Benchmark**	**Library Output**
Fig 1: WASP-76b 4.5 μm Phase Curve	May et al. (2021)	`May2021UltraHotPhaseCurveModel`
Fig 2: WASP-121b 4.5 μm Phase Curve	May et al. (2021)	`May2021UltraHotPhaseCurveModel`

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.